

Institute of Actuaries of India

Subject CT8 – Financial Economics

March 2017 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) In informational efficiency: The market for a particular security is said to exhibit informational efficiency if new information is incorporated quickly and accurately into the price of the security.

The major difficulties are:

- It is empirically difficult to determine exactly when a particular piece of information becomes available. When does the information become available to anyone (strong form efficiency) or publicly available (semi-strong form efficiency)?
- In order to test for strong form efficiency, you need access to information that is not publicly available.
- It can be difficult to decide exactly what constitutes publicly available information when testing the semi-strong form.
- It is difficult to judge exactly the extent to which the market price should react to a particular event and hence to determine whether or not it has in fact under- or over-reacted to that event. (This applies to all three forms.)

Empirical evidence concerning informational efficiency:

Many studies show that the market over- reacts to certain events and under- reacts to other events.

Over reaction to events

- Past winners tend to be future losers and the market appears to over-react to past performance.
- Certain accounting ratios appear to have predictive powers, an example of the market apparently over-reacting to past growth.
- Firms coming to the market have poor subsequent performance.

Under-reaction to events

- Stock prices continue to respond to earnings announcements up to a year after their announcement.
- Abnormal excess returns for both the parent and subsidiary firms following a demerger.
- Abnormal negative returns following mergers. (7)

ii) Efficient Market Hypothesis

In an efficient security market the price of every security fully reflects all available and relevant information. The EMH states that security markets are efficient.

a) This is inconsistent with the semi strong form of the EMH. If the semi strong form of EMH holds the low P/E should have an immediate effect on the share prices as the market should quickly and accurately respond to the information

b) The prices are reacting when information is made public. This suggests that the prices have previously been distorted by insider information. Therefore, this observation contradicts the strong form of EMH (4)

[11 Marks]

Solution 2:

i) Marginal utility of X i.e $MU_x = \frac{\partial U(X,Y)}{\partial X}$
 $0.4 X^{-0.6} Y^{0.6}$
 Marginal utility of Y i.e $MU_y = \frac{\partial U(X,Y)}{\partial Y}$
 $0.6 X^{0.4} Y^{-0.4}$

(2)

ii) $\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$
 $(0.4 X^{-0.6} Y^{0.6}) / 2 = (0.6 X^{0.4} Y^{-0.4}) / 6$
 $0.2Y = 0.1X$
 $X = 2Y$

(2)

iii) The consumer's budget constraint is
 $60 = 2X + 6Y$
 Substituting in the consumption for X in terms of Y we get

$$60 = 10Y \implies Y = 6 \text{ and } X = 12$$

(2)

iv) Suppose the consumer has total 100

Consider the scenario where an individual X has initial wealth of 100. He is given a gamble where there is 45% chance that he will lose 50 and 55% chance that he will gain 50.

Let utility function of the X be $U(W) = \sqrt{W}$

Suppose X takes the gamble. Then, expected wealth of X is

$$E(W) = 55\% (150) + 45\% (50) = 102.5$$

However, expected utility of X is

$$E(U(W)) = 55\% (\sqrt{150}) + 45\% (\sqrt{50}) = 9.92$$

Initial expected utility of X = 10 So, if X were to maximise expected wealth, he should take the gamble, but if he were to maximise expected utility, he should not take the gamble.

(4)

[10 Marks]**Solution 3:**

i) Given

$$F(t,x) = e^{-t}x^2$$

$$\frac{df}{dt} = -e^{-t}x^2 = -f$$

$$\frac{df}{dx} = 2e^{-t}x$$

$$\frac{df}{dx^2} = 2e^{-t}$$

$$Y_t = f(t, X_t)$$

Applying Ito's lemma

$$\begin{aligned} dY_t &= \frac{df}{dt} dt + \frac{df}{dx} dX_t + \frac{1}{2} \frac{d^2f}{dx^2} \sigma^2 X_t^2 dt \\ &= -fdt + 2e^{-t} X_t dX_t + e^{-t} \sigma^2 X_t^2 dt \\ &= -Y_t dt + 2e^{-t} X_t^2 \frac{dX_t}{X_t} + e^{-t} \sigma^2 X_t^2 dt \\ &= -Y_t dt + 2Y_t [0.25 dt + \sigma dW_t] + \sigma^2 Y_t dt \end{aligned}$$

$$\begin{aligned} \frac{dY_t}{Y_t} &= [2 * (0.25) - 1 + \sigma^2] dt + 2\sigma dW_t \\ &= [\sigma^2 - 0.5] dt + 2\sigma dW_t \end{aligned}$$

Therefore

$$dY_t = [\sigma^2 - 0.5] Y_t dt + 2\sigma Y_t dW_t$$

(4)

ii) The process is martingale if drift is zero. This means $\sigma^2 - 0.5 = 0$

i.e. $\sigma^2 = 0.5$

(1)

[5 Marks]

Solution 4:

i) Using Black-Scholes model to value the equity, we get:

$$\begin{aligned} d1 &= [\ln(22,000,000/30,000,000) + (.06 + .39^2/2) * 10] / (.39 \sqrt{10}) \\ &= .8517 \\ d2 &= .8517 - (.39 \sqrt{10}) \\ &= -.3816 \end{aligned}$$

$$N(d1) = .8028$$

$$N(d2) = .3514$$

Putting these values into Black-Scholes:

$$\begin{aligned} \text{Value of Equity} &= 22,000,000(.8028) - (30,000,000e^{-.06(10)}) (.3514) \\ &= \text{INR } 11,876,514.69 \end{aligned}$$

(3)

ii) The value of the debt is the company's value minus the value of the equity,

$$D = 22,000,000 - 11,876,514.69$$

$$= \text{INR } 10,123,485.31$$

(1)

iii) Using the equation for the PV of a continuously compounded lump sum, we get:

$$10,123,485.31 = 30,000,000e^{-R(10)}$$

$$.33745 = e^{-R10}$$

$$R_D = -(1/10)\ln(.33745)$$

$$= 10.86\%$$

(2)

iv) Using Black-Scholes model to value the equity, we get:

$$d1 = [\ln(22,750,000/30,000,000) + (.06 + .39^2/2) * 10]$$

$$(.39 * \sqrt{10})$$

$$= .8788$$

$$d2 = .8788 - (.39 \sqrt{10})$$

$$= -.3544$$

$$N(d1) = .8103$$

$$N(d2) = .3615$$

Putting these values into Black-Scholes:

$$E = \text{Rs. } 22,750,000(.8103) - (\text{Rs. } 30,000,000e^{-.06(10)})(.3615)$$

$$= \text{INR } 12,481,437.06$$

(2)

v) The value of the debt is the company's value minus the value of the equity, so:

$$D = 22,750,000 - 12,481,437.06$$

$$= \text{INR } 10,268,562.94$$

Using the equation for the PV of a continuously compounded lump sum, we get:

$$10,268,562.94 = 30,000,000e^{-R(10)}$$

$$.35429 = e^{-R10}$$

$$R_D = -(1/10)\ln(.35429)$$

$$= 10.72\%$$

When the firm accepts the new project, part of the NPV accrues to bondholders. This increases the present value of the bond, thus reducing the return on the bond. Additionally, the new project makes the firm safer in the sense that it increases the value of assets, thus increasing the probability the call will end in-the-money and the bondholders will receive their payment.

(2)

[10 Marks]

Solution 5:

i) Assumptions – Stock price follow geometric Brownian motion

$$dS_t = [\mu dt + \sigma dZ_t] S_t$$

Risk free rate, volatility, is assumed to be constant

$$(dS_t)^2 = [\mu^2(dt)^2 + \sigma^2(dZ_t)^2 + 2\mu\sigma dt dZ_t] S_t^2$$

$$= [\mu^2(dt)^2 + \sigma^2(dt) + 2\mu\sigma dt dZ_t] S_t^2$$

$$= \sigma^2 S_t^2 (dt)$$

Using these results and Ito's lemma, we get

$$df(S_t, t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} (dS_t)^2$$

$$= \Theta dt + \Delta dS_t + \frac{1}{2} \Gamma \sigma^2 S_t^2 (dt)$$

Now construct the portfolio

Minus 1 derivative

Plus Δ shares (0.5)

Value of this portfolio is such that

$$\begin{aligned} dV_t &= -df(S_t, t) + \Delta (dS_t + q S_t dt) \\ &= -\left\{ \Theta dt + \Delta dS_t + \frac{1}{2} \Gamma \sigma^2 S_t^2 (dt) \right\} + \Delta (dS_t + q S_t dt) \\ &= -\left\{ \Theta - q \Delta S_t + \frac{1}{2} \Gamma \sigma^2 S_t^2 \right\} (dt) \end{aligned}$$

We now notice that this term does not involve dZ_t and hence it is non – random, depending only on the change in time dt .

By the principle of no arbitrage this portfolio must earn risk free rate of interest

$$\text{ie. } dV_t = r V_t dt$$

Putting these two equations together we get

$$-\left\{ \Theta - q \Delta S_t + \frac{1}{2} \Gamma \sigma^2 S_t^2 \right\} (dt) = r V_t dt = r (-f(S_t, t) + \Delta S_t) dt$$

$$r f(S_t, t) = \Theta + (r-q) \Delta S_t + \frac{1}{2} \Gamma \sigma^2 S_t^2$$

Which is the Black Scholes PDE (4)

- ii) The delta of the call option is $N(d1)$ and the delta of the put option is $N(d1) - 1$. Since you are selling a put option, the delta of the portfolio is $N(d1) - [N(d1) - 1]$. This leaves the overall delta of your position as 1. This position will change dollar for dollar in value with the underlying asset. This position replicates the dollar “action” on the underlying asset.

(2)

- iii) The delta indicates that when the value of the of the INR exchange rate increases by \$0.01, the value of the bank's position increases by
- $$0.01 * 50000 = 500$$

The gamma indicates that when the INR exchange rate increases by \$0.01 the delta of the portfolio decreases by $0.01 * 100,000 = 1000$ (2)

iv) For Delta Neutrality 50000 INR should be shorted (2)

v) When the exchange rate moves to 1.0, we expect the delta of the portfolio to decrease by $(1.0 - 0.9) * 100,000 = 10,000$ so that it becomes 40,000 (2)

vi) To maintain delta neutrality, it is therefore necessary for the bank to unwind its short position 10,000 INR so that a net 40,000 INR have been shorted. When a portfolio is delta neutral and has a -ve gamma, a loss is experienced when there is a large movement in the underlying asset price. Given this it can be concluded that the bank is likely to have lost money. (3)

[15 Marks]

Solution 6:

i) A recombining binomial tree or binomial lattice is one in which the

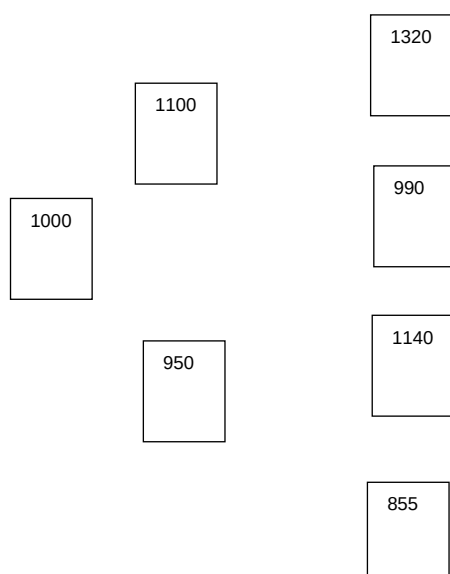
sizes of the up-steps and down-steps are assumed to be the same under all states and across all time intervals.

i.e., $u_t(j) = u$ and $d_t(j) = d$ for all times t and states j , with $d < \exp(r) < u$

• It therefore follows that the risk neutral probability 'q' is also constant at all times and in all states eg. $q_t(j) = q$

• The main advantage of a 'n' period recombining binomial tree is that it has only $[n+1]$ possible states of time as opposed to 2^n possible states in a similar non-recombining binomial tree. This greatly reduces the amount of computation time required when using a binomial tree model.

• The main disadvantage is that the recombining binomial tree implicitly assumes that the volatility and drift parameters of the underlying asset price are constant over time, which assumption is contradicted by empirical evidence



(4)

ii) a) The risk-neutral probabilities at the first and second steps are as follows:

$$q_1 = (\exp(0.0175) - 0.95)/(1.10 - 0.95)$$

$$= (0.06765)/0.15$$

$$= 0.4510$$

$$q_2 = (\exp(0.025) - 0.90)/(1.20 - 0.90)$$

$$= 0.41772$$

Put payoffs at the expiration date at each of the four possible states of expiry are 0,0,0 and 95.

Working backwards, the value of the option V1 (1) following an up step over the first 3 months is

$$V1(1) \exp(0.025) = [0.41772 * 0] + [0.58228 * 0]$$

$$\text{i.e., } V1(1) = 0$$

The value of the option V1 (2) following a down step over the first 3 months is:

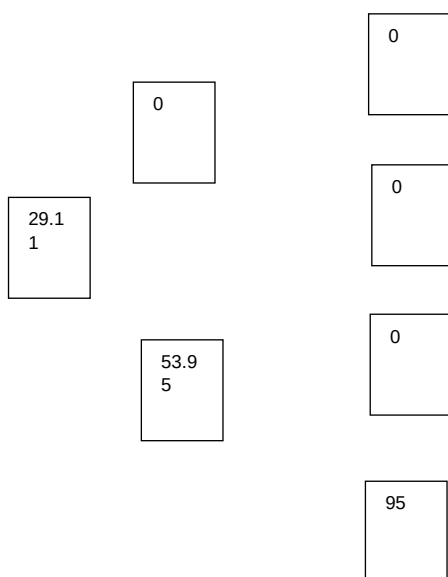
$$V1(2) \exp(0.025) = [0.41772 * 0] + [0.58228 * 95]$$

$$\text{i.e., } V1(2) = 53.9508$$

The current value of the put option is:

$$V0 \exp(0.0175) = [0.4510 * 0] + [0.5490 * 53.9508]$$

$$\text{i.e., } V0 = 29.105$$



(6)

b) While the proposed modification would produce a more accurate valuation, there would be a lot more parameter values to specify. Appropriate values of u and d would be required for each branch of the tree and values of ' r ' for each month would be required.

The new tree would have $2^6 = 64$ nodes in the expiry column. This would render the calculations prohibitive to do normally, and would require more programming and calculation time on the computer.

- An alternative model that might be more efficient numerically would be a 6-step recombining tree which would have only 7 nodes in the final column.

(4)

[14 Marks]

Solution 7:

i) The beta for a particular risk factor in a portfolio is the weighted average of the betas of the assets. This is true whether the betas are from a single factor model or a multi-factor model. So, the betas of the portfolio are:

$$B1 = .20(1.10) + .20(0.70) + .60(1.2)$$

$$B1 = 1.08$$

$$B2 = .20(0.75) + .20(1.50) + .60(-0.1)$$

$$B2 = 0.39$$

$$B3 = .20(0.30) + .20(-0.75) + .60(1.90)$$

$$B3 = 1.05$$

So, the expression for the return of the portfolio is:

$$R_i = R_f + 1.08 F_1 + 0.39 F_2 + 1.05 F_3 \quad (3)$$

$$\text{ii) } R_i = 3.5\% + 1.08 F_1 + 0.39 F_2 + 1.05 F_3$$

Which means the return of the portfolio is:

$$R_i = 3.5\% + 1.08(5.50\%) + 0.39(4.20\%) + 1.05(4.90\%)$$

$$R_i = 16.223\% \quad (1)$$

iii) The Vasicek model has the dynamics, under Q :

$$dr(t) = \alpha(\mu - r(t))dt + \sigma d\hat{W}(t)$$

where $\hat{W}(t)$ is a standard Brownian motion under Q

and the formula for bond price is:

$$B(t, T) = e^{a(\tau) - b(\tau)r(t)}$$

where

$$\tau = (T - t) = 5$$

$$b(\tau) = \frac{(1 - e^{-\alpha\tau})}{\alpha} = \frac{(1 - e^{-.1 \times 5})}{0.1} = 3.934693$$

$$\begin{aligned}
 a(\tau) &= (b(\tau) - \tau)\left(\mu - \frac{\sigma^2}{2\alpha^2}\right) - \frac{\sigma^2}{4\alpha} b(\tau)^2 \\
 &= (3.934693 - 5)\left(0.1 - \frac{.02^2}{2 \cdot 0.1^2}\right) - \frac{.02^2}{4 \cdot 0.1} 3.934693^2 \\
 &= (-1.065306) \cdot (.08) - .001 \cdot 15.481809 \\
 &= -.100706
 \end{aligned}$$

Therefore

$$\begin{aligned}
 B(t, t+5) &= e^{-0.100706 - 3.934693 \cdot 0.1} \\
 &= 0.610074
 \end{aligned}$$

The Cox-Ingersoll-Ross (CIR) model

The SDE for $r(t)$ under Q is:

$$dr(t) = \alpha(\mu - r(t))dt + \sigma \sqrt{r(t)} d\hat{W}(t)$$

Here the Std deviation is $\sigma \sqrt{r(t)} = .02$

$$\text{Therefore } \sigma = \frac{.02}{\sqrt{r(t)}} = \frac{.02}{\sqrt{0.1}} = .063245$$

In comparison with the Vasicek Model, the only difference is the volatility term, which is from the constant σ to $\sigma \sqrt{r(t)}$. The modification decreases the volatility of the short rate when the interest rate is low and thus the short rate is always positive. Hence the CIR model remedies the negative interest rate problem in the Vasicek Model.

For CIR Model the formula for bond price is:

$$B(t, T) = e^{a(\tau) - b(\tau)r(t)}$$

We start with

$$\theta = \sqrt{\alpha^2 + 2\sigma^2} = \sqrt{(.1)^2 + 2 \cdot (0.0632)^2} = 0.13416$$

Define

$$\begin{aligned}
 \beta &= (\theta + \alpha)(e^{\theta\tau} - 1) + 2\theta \\
 &= (.13416 + 0.1) \cdot (e^{0.13416 \cdot 5} - 1) + 2 \cdot 0.13416 \\
 &= 0.492130 \text{ (give credit for rounding)}
 \end{aligned}$$

$$\begin{aligned}
 b(\tau) &= \frac{2(e^{5\theta} - 1)}{\beta} \\
 &= 3.884345
 \end{aligned}$$

$$\begin{aligned}
 a(\tau) &= \frac{2\alpha\mu}{\sigma^2} \ln\left(\frac{2\theta e^{(\theta+\alpha)\tau/2}}{\beta}\right) \\
 &= \frac{2 \cdot 0.1 \cdot 0.1}{.063245^2} \ln\left(\frac{2 \cdot 0.13416 e^{(0.13416+0.1)5/2}}{0.492130}\right)
 \end{aligned}$$

$$= 5 * \ln \left(\frac{0.481825}{0.492130} \right)$$

$$= (-.105813)$$

$$\text{Therefore bond price} = e^{-0.105813 - 3.884335 * 0.1}$$

$$= 0.610031$$

(4)

[8 Marks]

Solution 8:i) Given $Z(t)$ is standard Brownian

$$\begin{aligned} \text{a. } dU(t) &= 2dZ(t) - 0 \\ &= 0dt + 2dZ(t). \end{aligned}$$

Thus, the stochastic process $\{U(t)\}$ has zero drift.

$$\begin{aligned} \text{b. } dV(t) &= d[Z(t)]^2 - dt. \\ d[Z(t)]^2 &= 2Z(t)dZ(t) + \frac{2}{2} [dZ(t)]^2 \\ &= 2Z(t)dZ(t) + dt \text{ by the multiplication rule} \end{aligned}$$

Thus, $dV(t) = 2Z(t)dZ(t)$. The stochastic process $\{V(t)\}$ has zero drift.

$$\begin{aligned} \text{c. } dW(t) &= d[t^2 Z(t)] - 2t Z(t)dt \\ \text{Because } d[t^2 Z(t)] &= t^2 dZ(t) + 2tZ(t)dt, \text{ we have} \\ dW(t) &= t^2 dZ(t). \end{aligned}$$

Thus The process $\{W(t)\}$ has zero drift

(9)

ii) a) Expected growth rate

$$\begin{aligned} \mu_V &= r_f + \beta(\mu_M - r_f) \\ &= 0.05 + 1.5(0.06) \\ &= 0.14 \end{aligned}$$

(1)

b) We can compute

$$\begin{aligned} \alpha_V &= \mu_V - \sigma^2 V / 2 = 0.14 - 0.302 / 2 = 0.095 \\ (\alpha_V T - \ln(F/V_0)) / \sigma_V \sqrt{T} \\ &= [0.095 - \ln(60/100)] / 0.30 \\ &= 2.0194 \end{aligned}$$

$$P(\text{Default}) = N(-2.0194) = 2.17\%$$

(4)

[14 Marks]

Solution 9:

i) Variance of return on security i as per single index model is given as

$$V_i = \beta_i^2 V_M + V\epsilon_i$$

$$\text{i.e. } \sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma^2 \epsilon_i$$

$$\sigma_A^2 = (.08^2 * .20^2) + .25^2 = 0.0881$$

$$\sigma_B^2 = (1.0^2 * .20^2) + .10^2 = .05$$

$$\sigma_C^2 = (1.2^2 * .20^2) + .20^2 = .0976$$

(3)

ii) If there are infinite number of assets with identical characteristics, then a well-diversified portfolio of each type will have only systematic risk since the non-systematic risk will approach zero with large n:

$$\text{Well diversified } \sigma_A^2 = (.08^2 * .20^2) + 0^2 = .0256 \quad \text{-----(I)}$$

$$\text{Well diversified } \sigma_B^2 = (1.0^2 * .20^2) + 0^2 = .04 \quad \text{----- (II)}$$

$$\text{Well diversified } \sigma_C^2 = (1.2^2 * .20^2) + 0^2 = .0576 \quad \text{----- (III)}$$

The mean will equal to that of the individual (identical) stocks.

$$E_A = 10\%$$

$$E_B = 12\%$$

$$E_C = 14\%$$

a) $E_A = 10\%, \sigma_A^2 = 0.0256$

(1)

b) Portfolio consists 55% of Type B

45% of Type C

Return on the portfolio

$$R_p = \sum x_i R_i$$

$$E(R_p) = E(\sum x_i R_i) = \sum x_i E(R_i) = .55 * 12\% + .45 * 14\%$$

$$= 12.9\%$$

$$\text{Var}(R_p) = \beta_p^2 V_M + V\epsilon_p$$

In case of well diversified portfolio $V\epsilon_p = 0$

$$\& \beta_p = x_i * \beta_i = .55 * \beta_B + .45 * \beta_C = .55 * 1.0 + .45 * 1.2 = 1.09$$

$$\text{Var}(R_p) = 1.09^2 * (\sigma_M^2)$$

$$= 1.09^2 * (.02)^2$$

$$= .047524$$

(2)

c) Arbitrage is the risk free trading profit. In this market, there is no arbitrage opportunity because on a well diversified portfolio all plot on the security market line (SML).

Because they are fairly priced, there is no arbitrage.

(2)

[8 Marks]

Solution 10:

i) a) Credit Risk:

Credit risk is the risk of loss due to contractual obligations not being met (in terms of quantity, quality and timing) either in part or in full, whether due to inability of, or decision by the counterparty. (1)

b) Credit spread:

Credit spread is a measure of the difference between the yield on a risky and a risk-free security, typically a corporate bond and a government bond respectively. (1)

ii)

In the Merton model, the company is modelled as having a fixed debt, L and variable assets F_t . This means the equity holders can be regarded as holding a European call on assets within the strike of L . It follows from the Black –Scholes model that we can deduce the (risk-neutral) default probability from the share price.

In the Black- Scholes formula for the price of call option, $\Phi d(2)$ represents the risk neutral probability that the option can be exercised or equivalently the risk neutral probability that the share price at expiry exceeds the strike price. Under the Merton Model approach, the call option replicates the shareholders' position and is exercised if shareholders repay the bondholders in full.

(3)

[5 Marks]
